PROJECT OVERVIEW

Bola Live

An Indonesian-language football site with live scores, fixtures, and standings in Jakarta time (WIB), for fans who follow European, Asian, and Southeast Asian leagues and national teams.

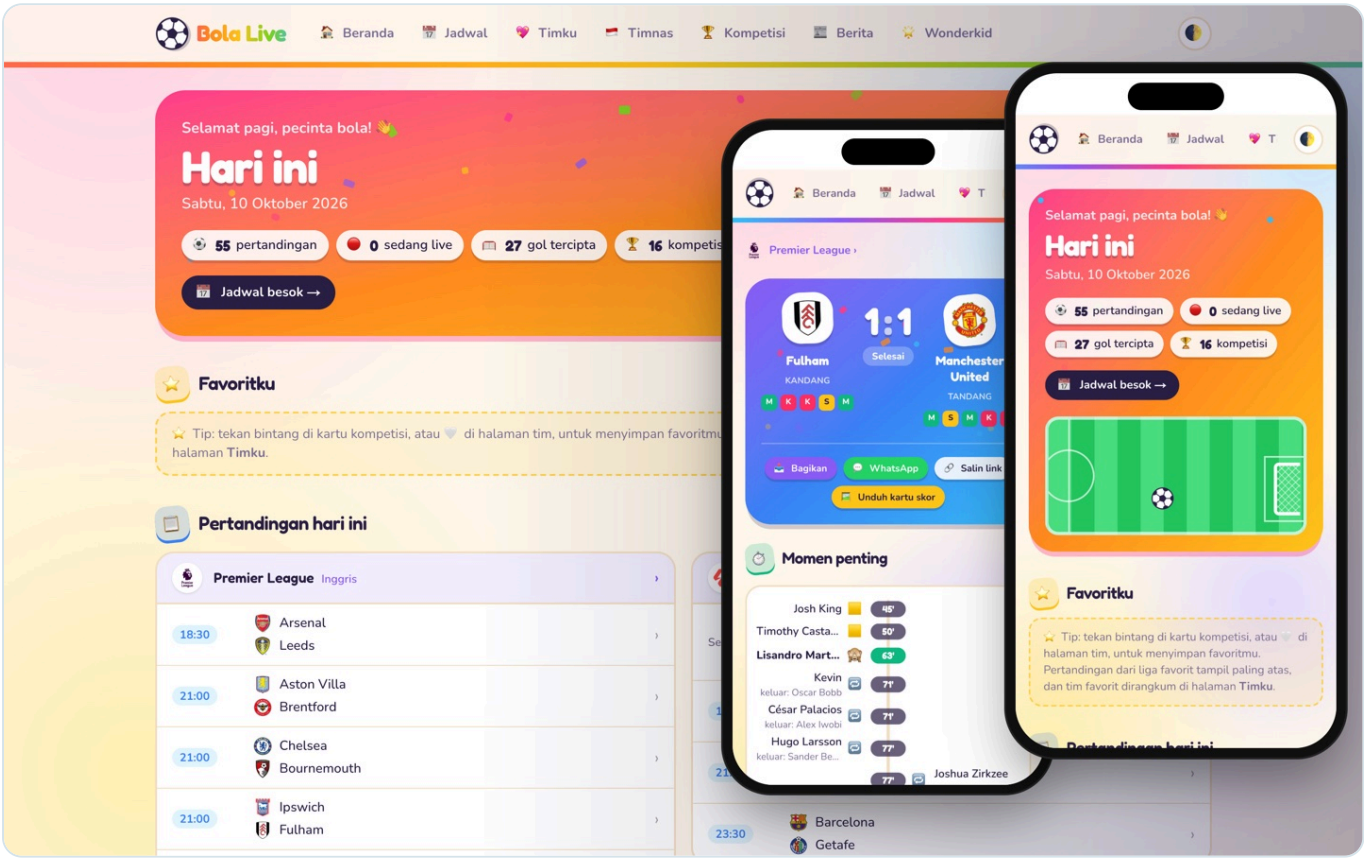
Raihan Evanza · Fullstack & Mobile Developer · portfolio-raihan-evanza.vercel.app

ROLE
Personal project · design, frontend, backend, deployment

PLATFORMS
Web (desktop & mobile), no account needed

LINKS
Live demo: bola-live-olive.vercel.app

Next.js 16 (App Router) · React 19 · TypeScript · Tailwind CSS 4 · ESPN API · TheSportsDB · Vercel



OVERVIEW

Popular football score sites are usually in English, show kick-off times in European or US time zones, and rarely cover Southeast Asian leagues. Fans in Indonesia end up converting times by hand and juggling several sites.

Bola Live brings 28 competitions together in one Indonesian-language site: the top 10 European leagues, the Champions/Europa/Conference League, the AFC Champions League, Asian and ASEAN leagues, and national-team tournaments. Every time is shown in WIB (UTC+7). Data comes from ESPN's public API, with ASEAN leagues from TheSportsDB.

The result is a Next.js site on Vercel with auto-updating live scores, match detail pages, team profiles, and per-device features such as favourite teams, score predictions, and goal notifications. All of it runs without a database or user accounts.

KEY FEATURES

Today's matches & live scores

The home page lists every match of the day by competition, with live matches pinned to the top and a summary of matches, live games, and goals. When a match is live or starts within 3 hours, the page pulls new scores every 30 seconds without a reload.

Fixtures by date & region

The fixtures page has a 7-day date strip and region filters (national teams, European club cups, top 10 European leagues, Asia, ASEAN). The selected date and region live in the URL, so the link can be shared.

Competition pages

Each competition has a standings table (including group/league phases and Champions League or relegation zones), top scorers and assists, the next 12 fixtures, and the last 12 results.

Match details

A match page shows key moments (goals, cards, substitutions), line-ups and formations, stats, minute-by-minute commentary, videos, and venue and referee info. Before kick-off it shows a countdown and buttons to add the match to Google Calendar or download an .ics file.

Club & national team profiles

Team pages show league position, last-five form, fixtures and results, the squad by position, and a club profile with history. The national-team page highlights Indonesia and has a search covering 193 countries.

Timku: favourites, predictions, goal alerts

Visitors can star competitions and teams, predict scores for upcoming matches (3 points for the exact score, 1 for the right result, locked at kick-off), and turn on browser notifications for their teams' goals. Everything is stored on the device, with no sign-up.

Shareable score cards

Every match has a 1200×630 PNG score card rendered on the server. It can be downloaded with the "download score card" button, next to the WhatsApp share and copy-link buttons.

FM26 wonderkids

A list of 43 sought-after young players in Football Manager 26, filterable by position, age, and league, with search by name, club, or country. The data is curated by hand.

ARCHITECTURE & DATA FLOW

Browser
Server-rendered pages; favourites, predictions, and theme kept in localStorage

→

Next.js (Vercel)
Server Components, route handlers /api/live, /api/timku, /api/tebakan, and score cards

→

Data Cache
Fetch results cached from 30 seconds to 24 hours, depending on the data

→

ESPN · TheSportsDB
Free public APIs, no key; ESPN for 24 competitions, TheSportsDB for 4 ASEAN leagues

- Pages are rendered on the server and regenerated periodically (ISR): the home and match pages every 30 seconds, competition pages every 60 seconds, team pages every 5 minutes. Visitors get ready-made HTML instead of calling ESPN from the browser.
- An AutoRefresh component calls router.refresh() every 30 seconds only while a match is live or about to start, and pauses while the tab is hidden.
- Per-device features (Timku, goal alerts, prediction scoring) send the team or match list from localStorage to small route handlers, which answer from the same cache.

STACK BY LAYER

Framework	Next.js 16 (App Router, Server Components, ISR) · React 19 · TypeScript
UI	Tailwind CSS 4 · Google Fonts (Fredoka, Nunito) · Light & dark mode
Data	ESPN public API · TheSportsDB (free tier) · Next.js Data Cache (fetch revalidate)
Browser features	localStorage + useSyncExternalStore · Notification API · Web Share API · Clipboard API
Images & calendar	next/og (ImageResponse) · Google Calendar link · File .ics
Deployment	Vercel

TECHNICAL DECISIONS

1. **Server-side caching tuned per data type**
   Every API request goes through Next.js fetch with a revalidate window: scores from yesterday to tomorrow 30 s, past days 6 h, future days 15 min, standings 5 min, top scorers 30 min, club profiles 24 h. However many people visit, ESPN is only called occasionally, which keeps load off an unofficial free API.

2. **WIB dates on top of US-based day boundaries**
   ESPN splits days by US time, so a 02:00 WIB kick-off lands on the previous day. For one WIB date the app fetches two ESPN days (D-1 and D), merges them, and filters with a fixed UTC+7 offset, which is safe because WIB has no daylight saving.

3. **One failing competition never breaks the page**
   Data for all 28 competitions is fetched in parallel with Promise.allSettled. Competitions that fail are listed in a small notice while the rest of the matches still render.

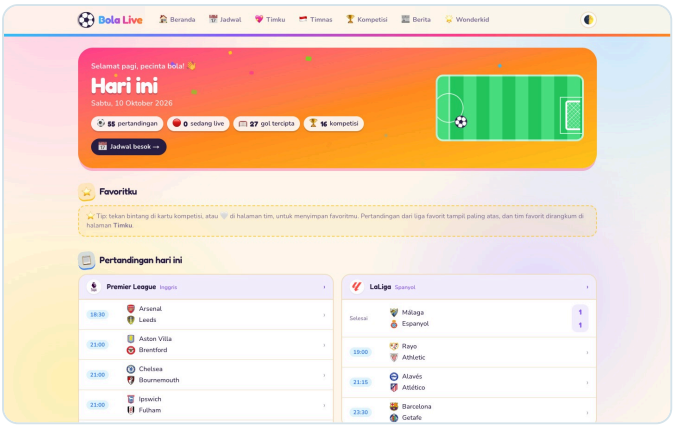
4. **Personal features without accounts or a database**
   Favourites, predictions, and theme live in localStorage behind a small useSyncExternalStore-based store. Every component updates when a value changes, even from another tab, and the server render uses defaults to avoid hydration mismatches. A tiny script in <head> applies the saved theme before first paint so dark mode does not flash.

5. **Lean refreshes and polling-based goal alerts**
   Auto-refresh only runs when a match is live or starts within 3 hours, and pauses while the tab is hidden. Goal alerts poll /api/live every 60 seconds; the first poll only records the baseline, then a notification fires when the goal total rises, a match kicks off, or it ends.

6. **A fallback source for ASEAN leagues**
   ESPN stopped updating the Indonesian, Thai, Malaysian, and Singaporean leagues, so any competition with a TheSportsDB ID is routed there instead. Its free tier allows roughly 30 requests per minute, so that data is cached for 10 minutes and labelled as limited in the UI.

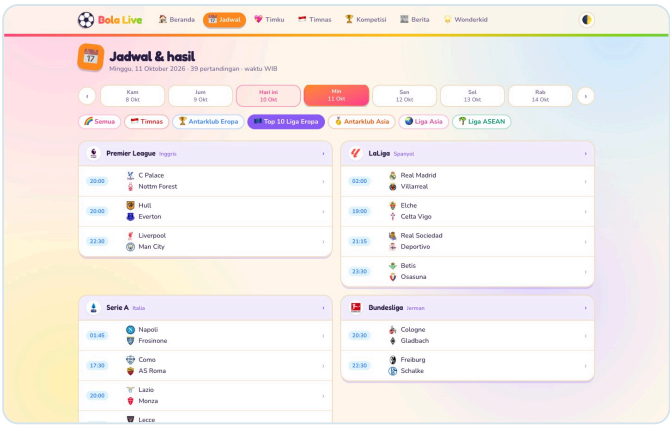
HOW TO USE IT

These steps follow the same order as the tutorial video.



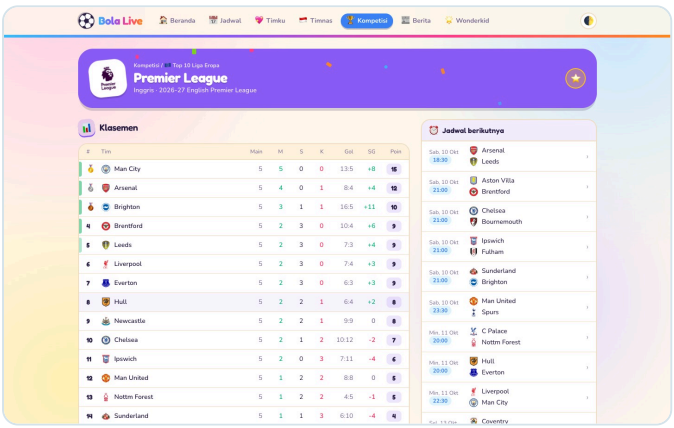
1 See today's matches on the home page

Open bola-live-olive.vercel.app. The top banner sums up today's matches, live games, and goals, and below it every match is grouped by competition with WIB kick-off times.



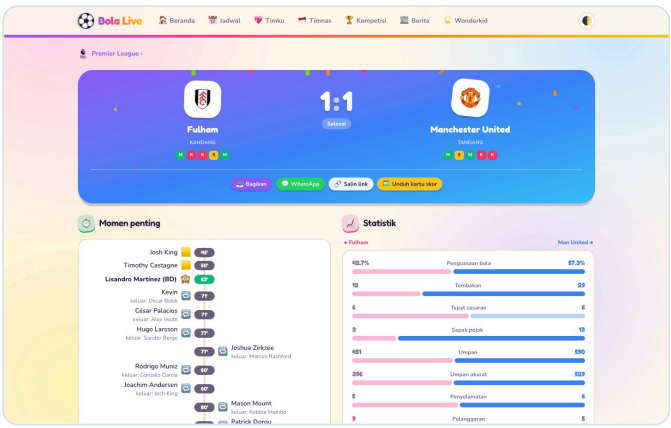
2 Pick a date and region in Jadwal

Click "Jadwal" (Fixtures), choose another day from the date strip, then click a region filter such as "Top 10 Liga Eropa" to narrow the list.



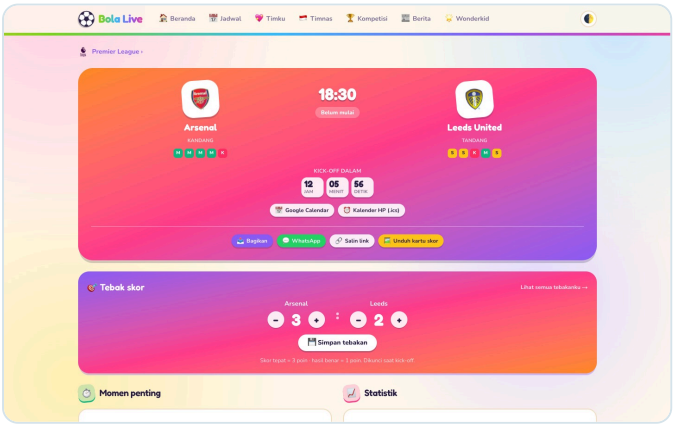
3 Open a competition table

Click "Kompetisi" (Competitions), tap the star on the Premier League card to favourite it, then open the card to see the table, top scorers and assists, upcoming fixtures, and recent results.



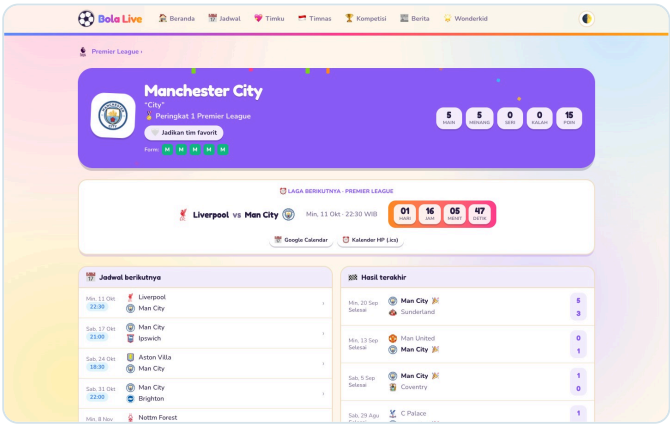
4 View a match in detail

Under "Hasil terakhir" (recent results), click a match. The detail page shows key moments, line-ups, stats, and minute-by-minute commentary, plus a "download score card" button.



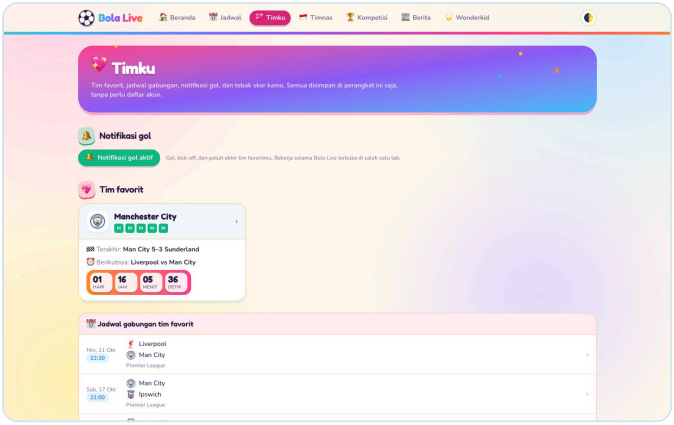
5 Predict an upcoming score

Go back to the competition and open a match under "Jadwal berikutnya" (upcoming). Set a score with the + and – buttons and press "Simpan tebakan" (save); points are scored automatically after full time.



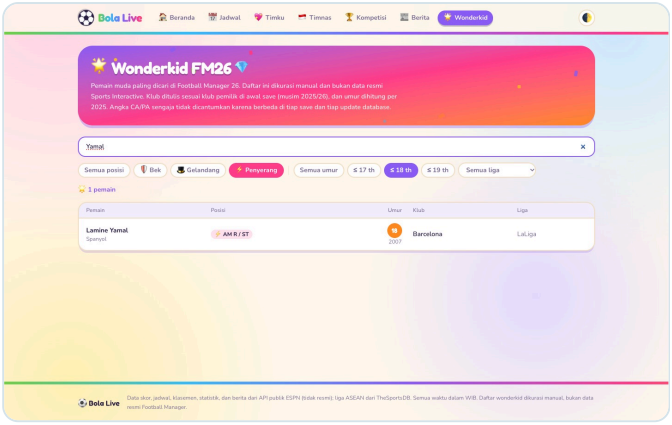
6 Save a favourite team

Click a team name in the table to open its profile (form, squad, fixtures, history), then press "Jadikan tim favorit" (make favourite).



7 Check your summary in Timku

Click "Timku" (My teams). Your favourite team is summarised with form, last result, and fixtures, next to your score predictions. Goal notifications can be switched on here too.



8 Browse FM26 wonderkids

Click "Wonderkid", pick the "Penyerang" (forward) position or an age limit, then type a player name into the search box.

LIMITATIONS & NOTES

- ESPN's API is unofficial and undocumented, so its structure can change without notice. This is a portfolio project, not for commercial use.
- Scores are not real time: with a 30-second cache and a 30-second refresh, what is on screen can trail the pitch by about a minute.
- ASEAN leagues use TheSportsDB's free tier: no live scores, stats, or timelines, and the number of matches per day and standings rows shown are capped.
- Korean, UAE, Qatari, and Czech leagues aren't available on ESPN, so they're not included. News and match commentary stay in English, as provided by the source.
- Goal alerts only work while Bola Live is open in a tab; there is no server push yet. Favourites and predictions are stored per device and do not sync.
- The wonderkid list is hand-curated, not official Sports Interactive data, and CA/PA ratings are deliberately left out.

WHAT'S NEXT

- Web Push via a service worker so goal alerts arrive even when the site is closed.
- Optional sign-in to sync favourites and predictions across devices, plus a prediction leaderboard.
- Automated tests for the ESPN data mapping and WIB date handling.